C U Vivek

Senior Performance & Systems Engineer

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Senior Performance and Systems Engineer with 9+ years of experience designing, benchmarking, and optimizing large-scale distributed systems in cloud-native environments. Proven expertise in end-to-end performance engineering, root cause analysis, and system scalability. Strong background in defining KPIs, building performance automation frameworks, and driving cross-functional initiatives to enhance system reliability and engineering excellence at scale.

PROFESSIONAL EXPERIENCE

Senior Performance Engineer, American Express June 2021 - Present
Bengaluru, India

- Lead performance and system engineering initiatives for large-scale Payment Network Authorization platforms operating under high-throughput production workloads.
- Designed and executed benchmarking strategies validating 6,000+ TPS transaction throughput under peak-scale conditions.
- Performed deep root cause analysis of latency spikes and throughput degradation using JVM profiling, heap/thread dump analysis, and Kafka consumer lag diagnostics.
- Architected and implemented a Shift-Left Performance Framework integrated with GitHub Actions, enabling automated performance validation in CI/CD and significantly reducing post-production performance defects.
- Conducted performance validation of Couchbase clusters under high concurrency workloads, optimizing index design and memory configuration to improve query latency and cluster stability.

Performance Engineer, Amdocs India LLC Nov 2019 - June 2021
Pune, India

- Part of Microservices-based Digital Application NFT Testing Team in Amdocs Core (R&D division). Agile-based monthly drop release Application Testing and Certification on various platforms like Openshift (Kubernetes), AWS EKS environment.
- Gathered non-functional requirements from product stakeholders and translated them into measurable performance benchmarks.
- Analyzed CPU, memory, and I/O utilization patterns to identify scaling thresholds and capacity limits.
- Monitored Kafka messaging performance and system behavior under distributed load conditions.

Software Test Engineer, Petrolink Data Services July 2019 - Oct 2019
Pune, India

- Developed and executed performance test scripts using LoadRunner for web and web service protocols.
- Supported execution environments and resolved load generation and performance infrastructure issues.

Software Test Engineer, Accenture Solutions Dec 2016 - June 2019
Pune, India

- Contributed to application testing and performance validation activities in enterprise environments.

CERTIFICATIONS

AWS Certified Solutions Architect Associate(SAA)

AWS Certified Cloud Practitioner - (CLF)

Microsoft Certified: Azure Fundamentals

TECHNICAL SKILLS

Programming & Automation -
Python, Shell Scripting

Distributed Systems -
Kafka, Microservices Architecture
Cloud & Container Platforms
AWS, EKS, Kubernetes, OpenShift

Performance Tools -
JMeter, K6, LoadRunner

Observability & Profiling -
Prometheus, Grafana, Dynatrace, Heap and Thread Dump Analysis

CI/CD & DevOps -
GitHub Actions, Jenkins

AWARDS

Centurion Award 2025
American Express
Awarded to top 1% Technology colleagues for their outstanding achievements

EDUCATION

B. Tech from VIT Vellore
2012 - 2016
CGPA - 8.86

KEY IMPACT

- Scaled distributed payment systems to 6,000+ TPS with stable latency under peak conditions.
- Reduced post-production performance incidents through CI-integrated benchmarking and shift-left validation.
- Improved system resiliency through structured chaos testing and failover validation across applications.